DATA SCIENCE CAPSTONE SHOWCASE

Predicting Contraceptive Use Among Kenyan Women

A Machine Learning & Deep Learning Approach to Targeting Family Planning Interventions

Prepared by: The Insight Architects

Joy Nyuguto • Martin Kitema • Shadrack Symekah • Clare Simiyu • Antony Sila • David Theuri

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Exploratory Data Analysis

Understanding the demographic, socioeconomic, and geographic landscape

Project Overview & Target Distribution

The Core Problem

Kenya's modern contraceptive prevalence rate (mCPR) is ~60% nationally, but this hides severe regional and socioeconomic disparities.

Our goal: predict modern contraceptive use to help health planners target resources effectively.

Target Variable (4-Class)

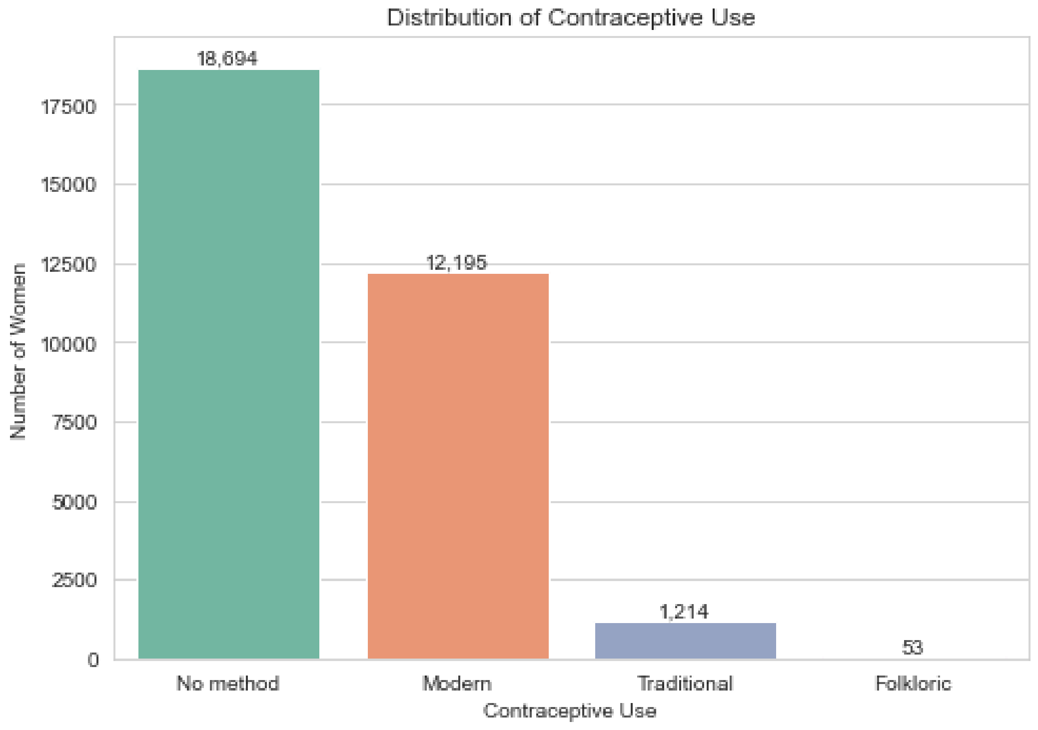
No method: 58.1% of women

Modern method: 37.9%

Traditional: 3.8%

Folkloric: 0.2%

The extreme class imbalance (0.2% Folkloric) is a key modeling challenge addressed via SMOTE resampling.



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Demographics: Age Distribution

Age Profile

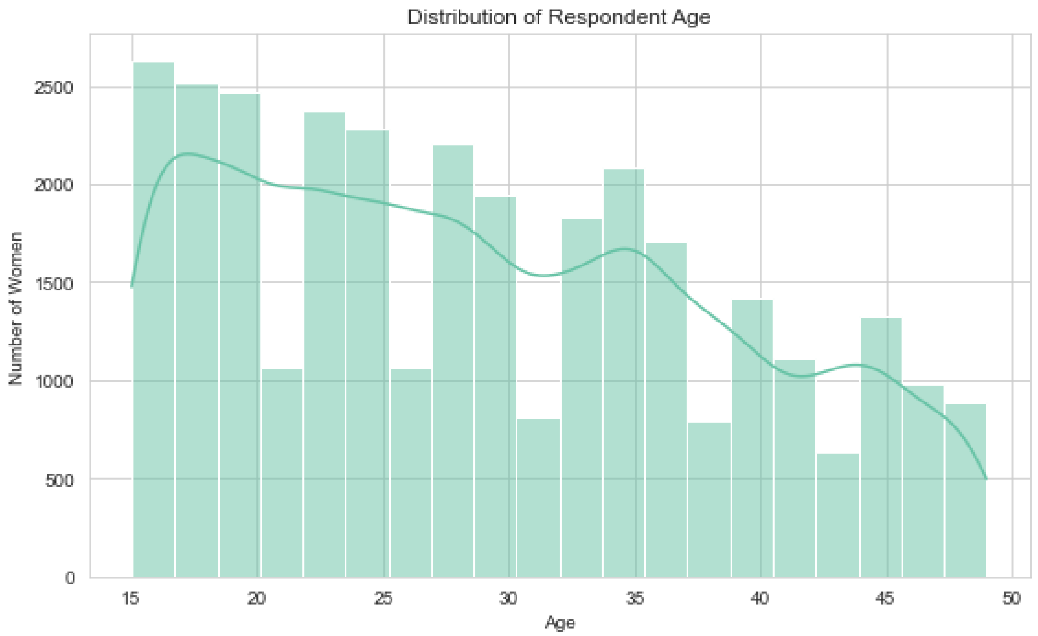
Most women are in their twenties (mean ≈ 29, median 28). Numbers thin toward the upper bound of 49.

This captures the primary reproductive age demographic accurately (KDHS 2022 sampled women aged 15–49).

Initial Numeric Summary

Household size and living children are heavily right-skewed.

```
print('Rows and columns:', df.shape)
df.describe().T
```



Demographics: Education Level

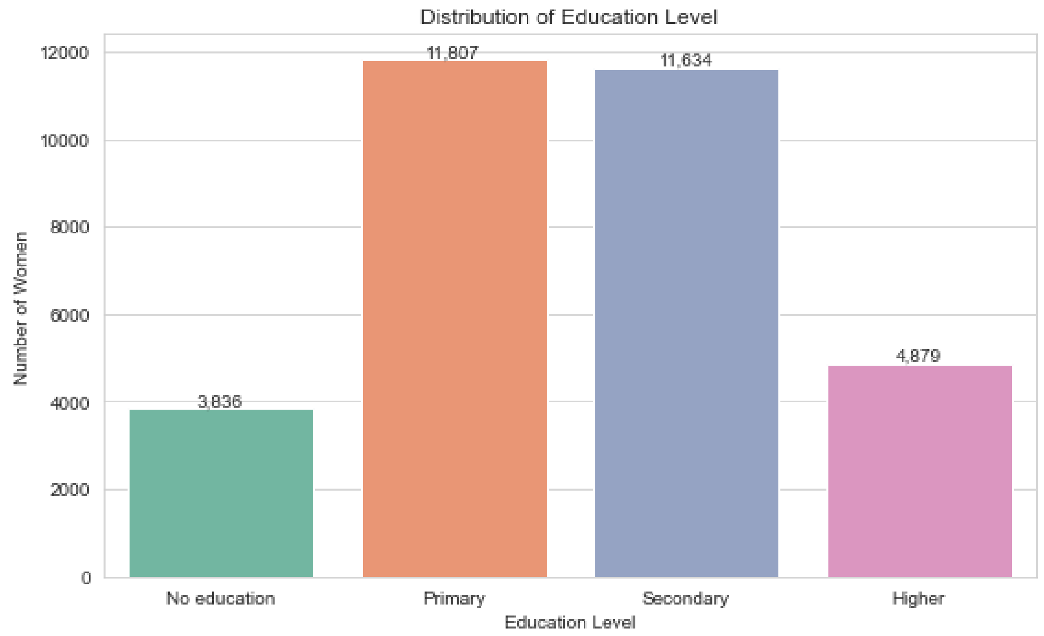
Education Level Distribution

Primary & Secondary: ~37% each

Higher education: ~15%

No formal education: ~12%

A large share of the target demographic has only primary schooling or below, underscoring the need for literacy-linked health campaigns.



Socioeconomics: Wealth & Residence

Wealth Index

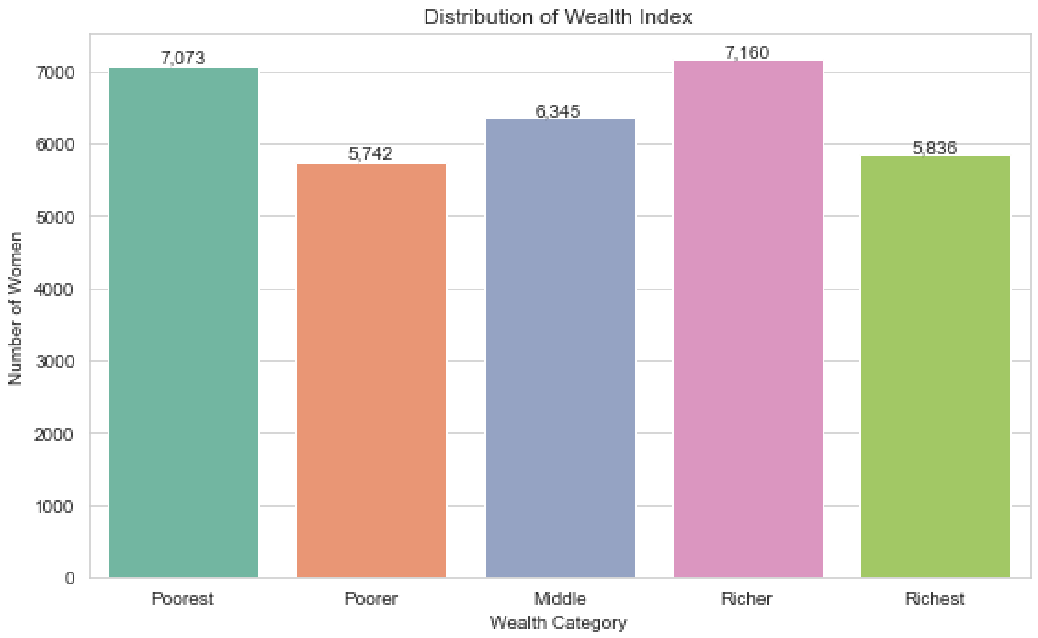
Five wealth quintiles are roughly balanced (18–22% each), giving even economic representation.

Residence Type

Rural: 62%

Urban: 38%

Urban women have marginally higher resource access, but the rural majority is the primary focus for policy.



Bivariate: Education as a Driver of Uptake

Education as a Catalyst

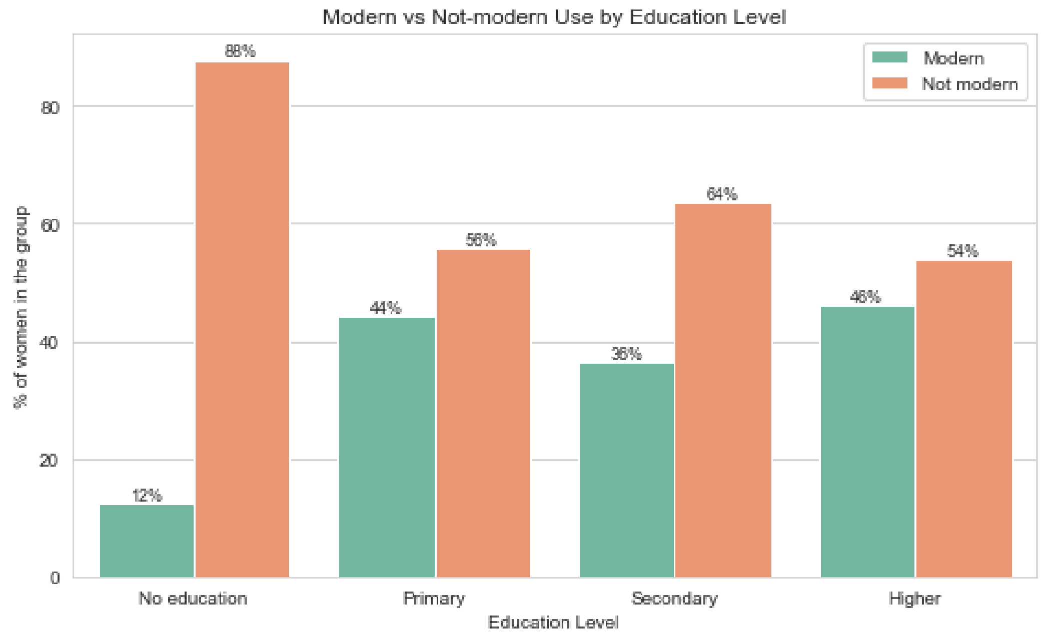
The starkest gap: women with no education (~12% modern use) vs. any education (~37–46%).

Even primary education significantly boosts adoption.

Wealth: A Threshold, Not a Gradient

The Poorest quintile stands out (~25% use). Above that, rates flatten at 41–43%.

Once extreme poverty is cleared, non-economic factors dominate.



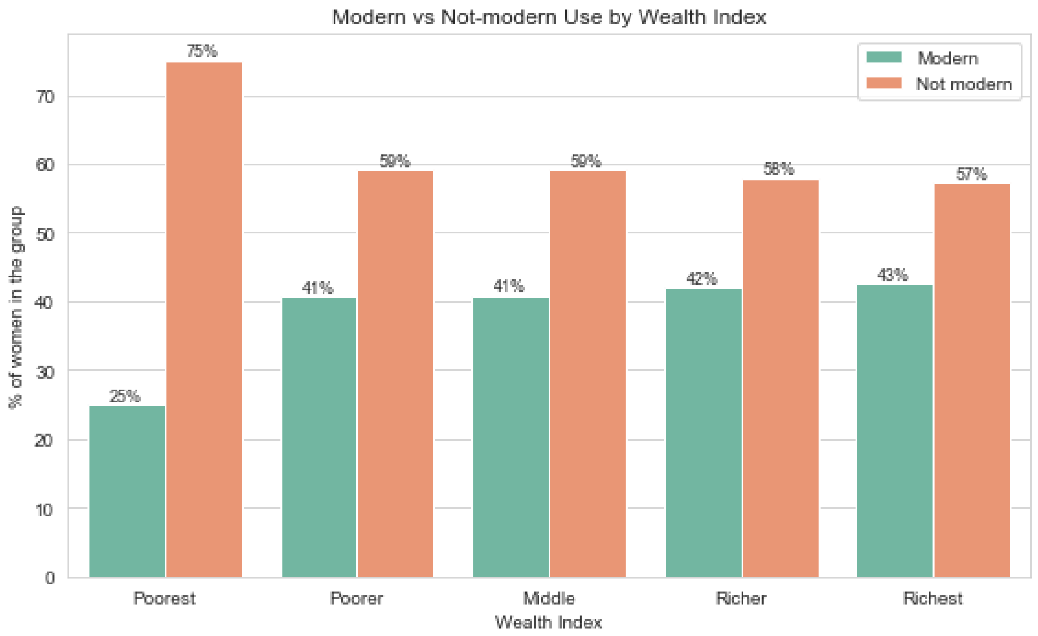
Bivariate: Wealth as a Driver of Uptake

Socioeconomic Barriers

The Poorest quintile is a critical cliff where use drops to ~25%.

This indicates a barrier requiring direct subsidies or free distribution to bridge.

Above the poverty line, uptake is stable (~42%), showing that education and geography matter more than incremental wealth.



Geographic Disparities: Modern Use by County

Massive County-Level Variation

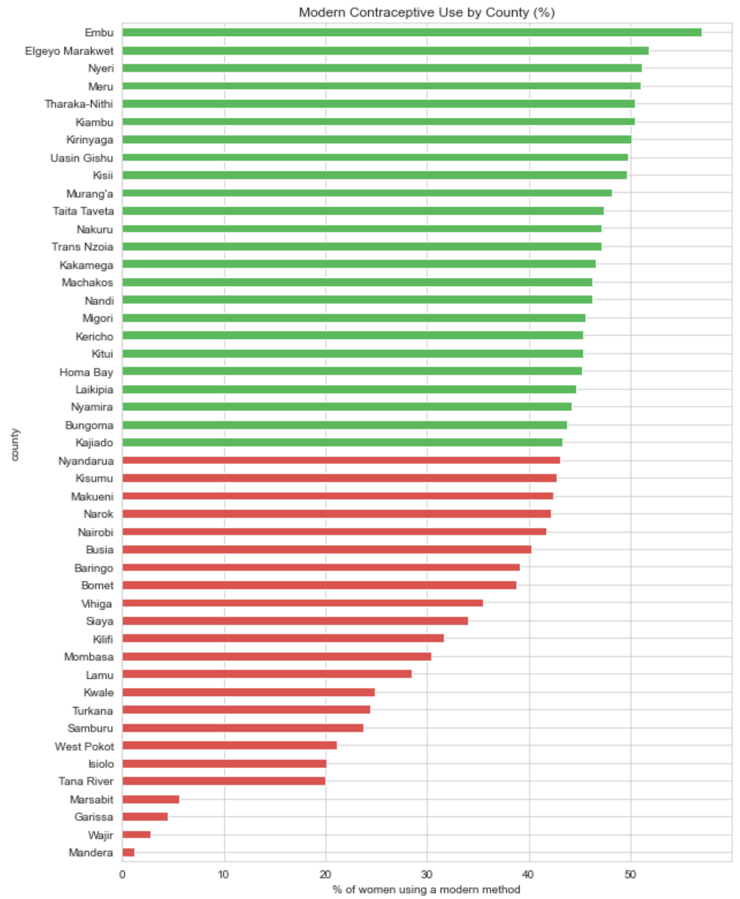
Lowest: Mandera (~1%), Wajir, Garissa, Marsabit (<6%)

Highest: Embu, Meru, Nyeri, Kirinyaga (>50%)

All low-use counties are in the arid/semi-arid north (ASAL regions), pointing to systemic infrastructure barriers beyond individual choice.

Implication

Geographically targeted interventions are essential — national averages mislead.



Multivariate: Does Wealth Overcome Lack of Education?

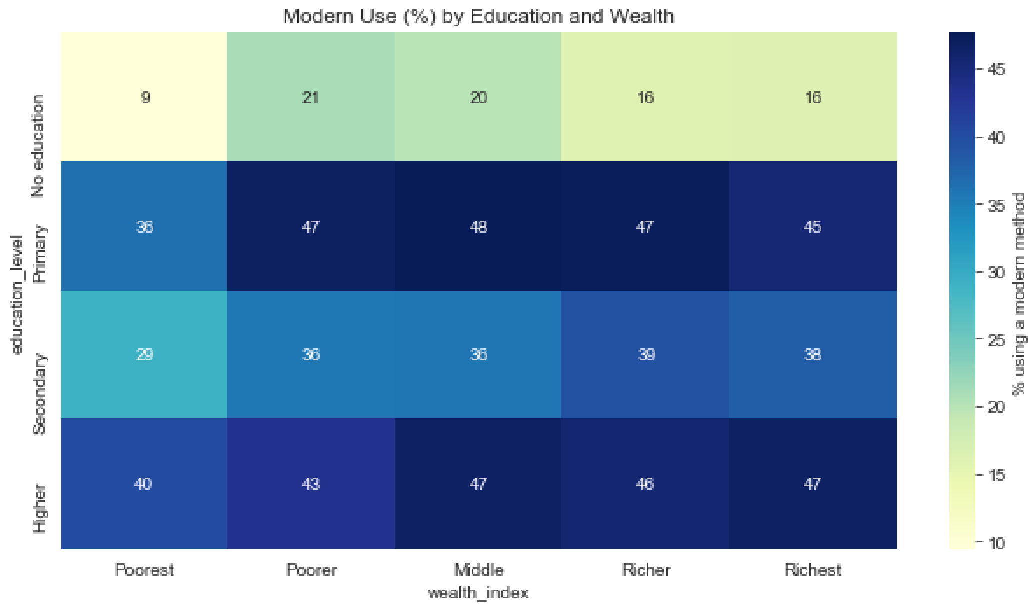
Education Trumps Wealth

Among women with no education, modern use stays 9–21% regardless of household wealth.

Conversely, highly educated women in the Poorest households still adopt at ~36%.

Key Takeaway

Literacy and health education are far more critical than income for long-term family planning adoption.



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Feature Correlation & Collinearity Audit

Correlation Matrix Findings

children_ever_born and living_children have r = 0.98 — nearly perfectly collinear.

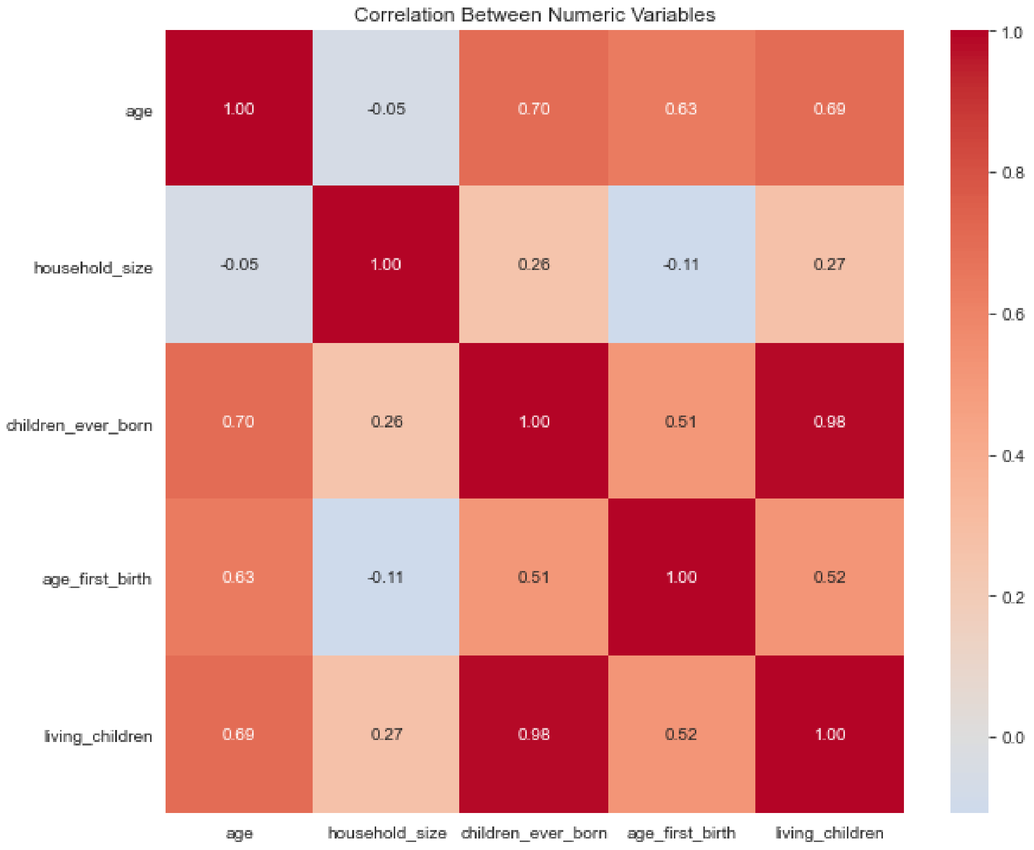
We resolved this by deriving child_loss and surviving_ratio instead.

Other Observations

Age correlates moderately with children ever born (~0.65)

Wealth and education show a moderate positive relationship (~0.45)

These informed our feature engineering decisions.



Data Engineering & Preprocessing

Cleaning, encoding, and preparing model-ready features

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Data Integrity & Cleaning Decisions

Structural Missingness

Two columns had missing values: age_first_birth and partner_education.

These are not errors — they represent women who have never given birth or have no partner.

Imputed 0 for age_first_birth for nulliparous women

Created a has_given_birth binary flag

Duplicates Resolution

274 rows flagged as duplicates. Audit confirmed they are distinct women with identical profiles.

All 32,156 rows deliberately retained for statistical integrity.

Post-Cleaning Validation

```
missing = df.isnull().sum()
print(missing[missing > 0])

print('Duplicate rows:', df.duplicated().sum())
```

Assertions Enforced

```
assert df.isnull().sum().sum() == 0
assert df.shape[0] == 32156
```

Every downstream step inherits guaranteed data completeness.

Feature Engineering: Encoding with Intent

Ordinal Encoding

Naturally ordered variables (education, wealth, age_group) encoded ordinally so tree models exploit monotonic structure.

Nominal One-Hot Encoding

County, region, religion, marital status → one-hot. Feature space: 36 → 126 columns.

Collinearity Resolution

Derived child_loss and surviving_ratio from the r=0.98 collinear pair.

Key Engineered Features

has_partner — 1 if partner education exists

education_gap — partner's minus woman's education

child_density — children / household size

arid_county — binary flag for ASAL regions

age_at_first_birth_gap — years since first birth

All features motivated directly by EDA findings.

Modeling & Evaluation

Five models benchmarked: from logistic regression to self-attention transformers

Resampling & Model Comparison

Train-Test Split & Resampling

Stratified 80/20 split preserves rare classes (Folkloric 0.2%, Traditional 3.8%).

SMOTE applied to training data only, upsampling minority classes to 14,955 rows each.

Selection Verdict

LightGBM (Tuned) selected for deployment: best Macro-AUC (0.798) and accuracy (72.4%).

Model	Accuracy	Macro F1	Macro AUC
LightGBM (Tuned)	**72.4%**	**0.367**	**0.798**
Random Forest	70.6%	0.390	0.775
Deep MLP	70.3%	0.385	0.751
TabTransformer	64.8%	0.384	0.754
Logistic Reg. (Base)	54.8%	0.354	0.764

Deep Learning Benchmarks

Deep MLP (4-Layer)

BatchNorm + Dropout + Adam optimizer.

70.3% accuracy, best Modern-class recall (75%)

AUC only 0.751 — discrete survey data limits gains

TabTransformer (Self-Attention)

13 categorical embeddings (16-dim) + 2-head attention.

34% recall on Traditional class (highest of all models)

LightGBM: 0%, Random Forest: 9%

Attention correctly captures region × religion × marital interactions.

TabTransformer Architecture

13 categorical embeddings (16-dim)

2-head self-attention layer

Concatenated with projected numerics

MLP classification head

Future Potential

Highly promising for production with more data. Attention weights identified confounding union-status and county structure.

Final Model Evaluation: Tuned LightGBM

Selected: LightGBM (Tuned)

Sequential gradient boosting; validation loss plateaued at iteration 226 (no overfitting).

Hold-Out Performance

Accuracy: 72.4%

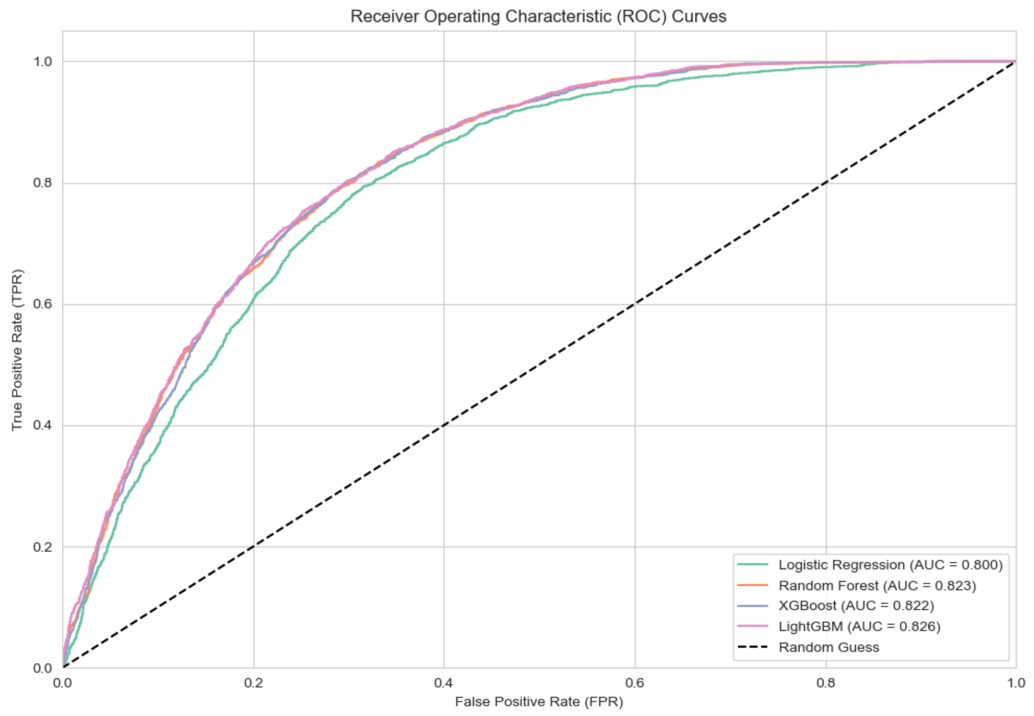
Macro-AUC: 0.798

Modern Recall: 71%

No-method Recall: 78%

Success Criteria

Planned targets (Acc ≥ 78%, AUC ≥ 0.82) not fully met due to the 4-class long-tail challenge. Realistic expectation-setting is essential in public health modeling.



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Confusion Matrix Analysis

Majority-Class Discrimination

No-method: 78% precision & recall

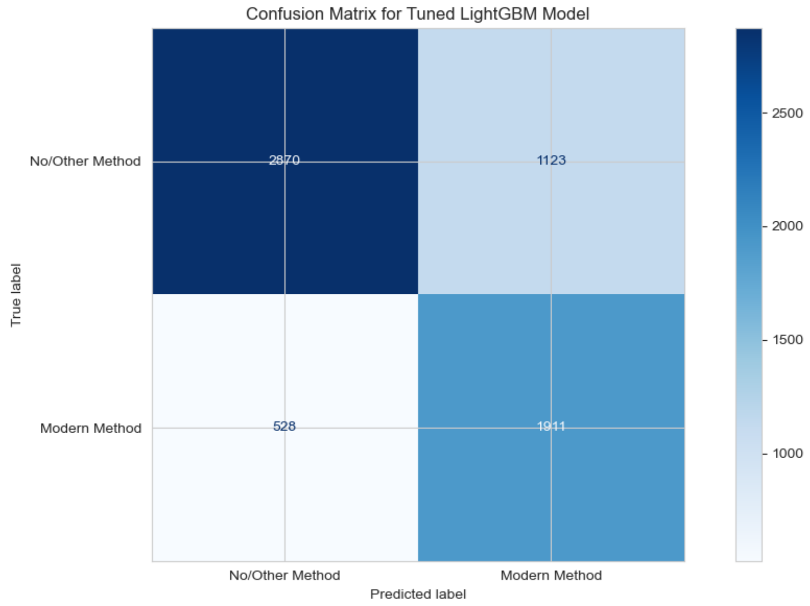
Modern: 64% precision, 71% recall

Rare-Class Suppression

LightGBM suppresses Traditional/Folkloric predictions. Gradient boosting downweights rare-class residuals once majority classes fit.

Actionable Insight

Deploy as a binary risk flag (likely non-user vs. likely modern user). 4-class predictions risk false precision with stakeholders.



Model Interpretability via SHAP

Top SHAP Drivers

1. Age

Strongest driver. Older women pushed toward modern use (spacing/limiting), younger away.

2. Living Children

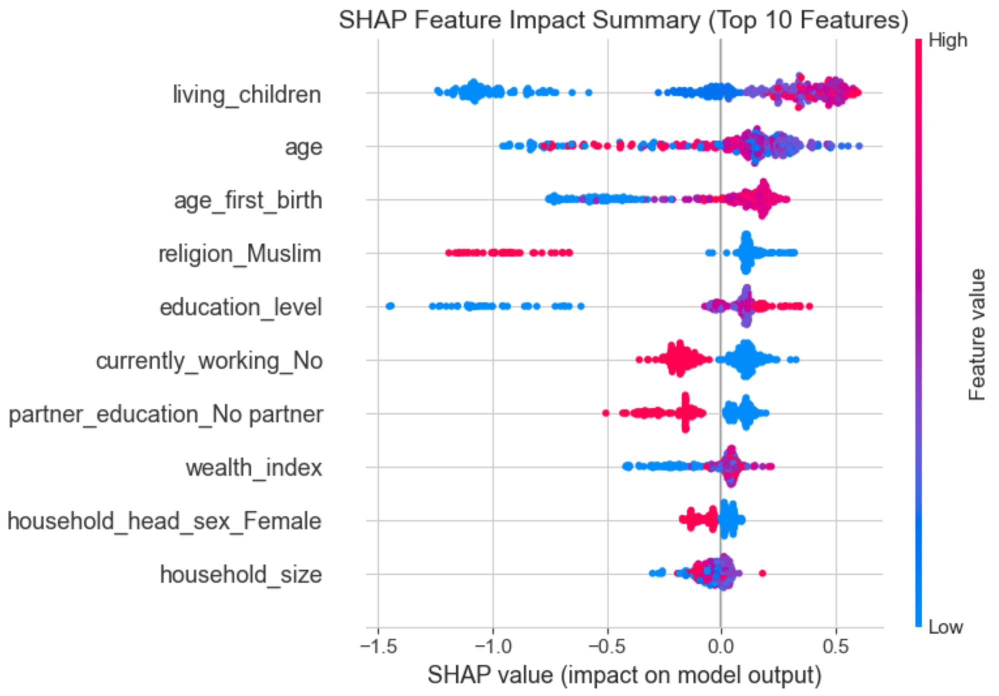
Women adopt contraception after having children, not before.

3. Employment

Most policy-actionable. Employed women pushed toward modern use.

4. Arid County

ASAL residence acts as a massive penalty regardless of wealth, education, or religion — pointing to infrastructure barriers.



Deployment & Recommendations

From model to actionable policy impact

Web Application & Serve-Skew Prevention

Interactive Web Application

Deployed on Render.com for healthcare stakeholders (Ministry of Health, Community Health Promoters).

Serve-Skew Prevention

Feature engineering code lives in a standalone feature_engineering.py module.

Both notebook and Flask API import the exact same file — mismatches structurally impossible.

Flask API Endpoints

```
GET  /         HTML Form Interface
GET  /health   Server Healthcheck
POST /predict  Single Prediction
```

Live Deployment

URL: kdhs-contraceptive-api.onrender.com

Auth: X-API-Key header required

Response: JSON with class probabilities

Actionable Policy Recommendations

1. Geographic Prioritization in ASAL Regions

Arid counties (Mandera, Wajir, Garissa) face systemic barriers pulling adoption below 6%. Fund mobile clinics and supply chains.

2. Target the Right Women

4-variable filter for Community Health Promoters:

Currently in a union

Unemployed

Poorest quintile

Living in an arid county

This isolates exposed non-users, avoiding wasted outreach visits.

3. Subsidize the Poorest Quintile

Modern use jumps from 25% (Poorest) to 41% (Poorer) then flattens. Target subsidies at the Poorest exclusively for maximum ROI.

Thank You!

Questions & Answers

GitHub:
github.com/symekah1999/Predicting-contraceptive-use-among-women

Live App:
kdhs-contraceptive-api.onrender.com